L'ORÉAL VeloCity AI

Selling Smarter: An Autonomous, Circular Sales Engine Turning Channel Decay into Protected Margin

A mobile-first edge AI architecture empowering 1,200+ on-ground Field Sales Officers (FSOs) to eliminate the 2.5% revenue bleed caused by expired, damaged, and stagnant beauty stock through 3-second shelf CV scans, predictive decay telemetry, and lateral B2B salon redistribution before inventory becomes landfill waste.

PILLAR 01 • SENSING

3-Sec Edge-CV Shelf Audit

On-device vision model (YOLOv8n) instantly quantifies shelf dust thickness, pack UV discoloration, and OCR batch expiry.

PILLAR 02 • INTELLIGENCE

Deterministic Tri-Band Engine

Automated inventory stratification: Green (Replenish), Amber (B2B Bundles), and Red (< 90d Expiry Immediate Swap).

PILLAR 03 • CIRCULARITY

Hyperlocal B2B StockSwap

Algorithmic lateral relocation to high-velocity professional salons within 3 km, zeroing central reverse freight.

CORE OPERATING ARCHITECTURE: FROM FORWARD PUSH TO CIRCULAR OFFTAKE

INTEGRATED WITH DISTRIBUTOR DMS & SAP S/4HANA

STAGE 01 • IN-STORE

Edge-CV Scan

FSO smartphone captures panoramic shelf audit & batch numbers in 3 seconds.

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STAGE 02 • THE BRAIN

Tri-Band Engine

Classifies DOI & expiry: 🟢 Green, 🟡 Amber (> 42d), 🔴 Red (< 90d).

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STAGE 03 • OFFTAKE

Lateral StockSwap

Hyperlocal matchmaker dispatches stock to nearby salon doors within 3 km.

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STAGE 04 • GOVERNANCE

Zero-Landfill Ledger

Instant DMS credit notes, tester conversions & gamified FSO Net Freshness bonuses.

₹42 Cr+	Annualized Protected EBITDA Margin
60% Cut	In Reverse Logistics & Destruction Costs
120 MT	Plastic Packaging Diverted From Regional Landfills
94 / 100	Territory Net Freshness Score (Up from 62)

The 2.5% Blindspot: Why Traditional FMCG Sales Bleed Downstream

Unpacking the structural drivers of channel decay across General Trade, Modern Trade, and Salons (Deliverable A)

PROBLEM & STRATEGY

ANATOMY OF THE 2.5% REVENUE BLEED

P&L DRAIN

Delivered & Consumed Revenue (Healthy Turnover)	97.5% (₹1,950 Cr)
Direct Write-Offs (Returns / Expiry / Damage)	~1.5% (₹30 Cr)
Reverse Freight, Central Handling & Incineration	~1.0% (₹20 Cr)

LEGACY FORWARD BLEED
Depot → Primary Booking → Shelf Dust (60d) → Central Depot Return → Landfill/Burn (₹50 Cr Drain)

VELOCITY CIRCULAR LOOP
Depot → CV Shelf Scan (30d) → Hyperlocal Lateral Swap (< 3km) → Full Value Salvage (+₹42 Cr Margin)

FRICTION POINT BREAKDOWN: LEGACY VS. VELOCITY AI

OPERATIONAL MODEL

Friction Point	Legacy SFA ("Push & Pray")	VeloCity AI Operating Model
Incentive Model	100% quota on gross order volume; zero penalty for downstream decay.	**Net Freshness Index (NFI):** 20% bonus tied to velocity and zero write-offs.
Shelf Visibility	Blind order-taking. Back-room dust and ageing stock ignored until return.	**3-Sec Edge-CV:** Auto-flags dust density, carton scuffs, and batch expiry.
Stagnant Stock	Reaction is to slash orders, triggering stock-outs on hero SKUs.	**B2B Dynamic Bundling:** Margin-optimized combos move Amber items.
Near-Expiry End	Depot return → central warehouse → destructive incineration.	**Lateral StockSwap:** Algorithmic transfer to nearby high-velocity doors.

EARLY DETECTION SIGNAL ARCHITECTURE (DELIVERABLE A)

FIELD PROTOCOL

01 Optical Dust Layer & UV Fading Density
Edge computer vision measures pixel luminance variation and carton micro-scuffs. Dust layer > 15 microns indicates > 45 days untouched shelf dwell time.
Trigger: Amber Warning flagged to FSO & Retailer instantly

02 Secondary Carton Deformation & Seal Wear
Identifies repeated customer handling without conversion (e.g. lipstick testers or hair treatment cartons opened, corner dog-ears, faded labels).
Trigger: Recommendation for 1-tap Salon Tester Conversion

03 Planogram Dwell & Displacement Drift
Detects when retailers push slow shades to bottom racks or behind hero products. System measures spatial coordinate shift vs. approved planogram.
Trigger: Guided AR re-anchoring to eye-level Golden Zone (1.2m)

04 Retailer Credit Drag & Velocity Delinquency
Cross-references retailer payment hesitation with Days-of-Inventory (DOI > 42 days). Catches stagnant batches before the retailer demands a credit return.
Trigger: Automated B2B Margin Bundle injection in distributor cart

180-SECOND IN-STORE FSO AUDIT PROTOCOL & ESCALATION WORKFLOW

FIELD-TESTED OPERATIONAL PROCESS

1. Store Check-In	2. Shelf Sweep	3. Edge Inference	4. Action Dispatch	5. DMS Sync
GPS Geofence Verified (15s)	Panoramic CV Capture (30s)	Bounding Box & OCR (45s)	Swap / Bundle Approval (60s)	Auto Credit Note (30s)

L'Oréal VeloCity AI: 3-Second Edge-CV Shelf Audit & Triage Engine

Autonomous on-ground detection pipeline running locally on the field sales representative's smartphone (Deliverables A & B)

AI & EDGE ARCHITECTURE

EDGE-AI DETECTION PIPELINE

OFFLINE-FIRST ON DEVICE

1 **3-Second Panoramic Shelf Scan**
FSO performs a single continuous camera sweep. A quantized on-device YOLOv8-nano model operates at 30 fps to segment facing counts, verify planogram alignment, and compute dust reflectivity score.

2 **OCR Batch & Expiry Harvester**
Integrated computer vision OCR extracts batch numbers, manufacturing dates, and lot stamps directly from secondary cartons or shelf tags, calculating exact Days-to-Expiry (DTE) with zero manual input.

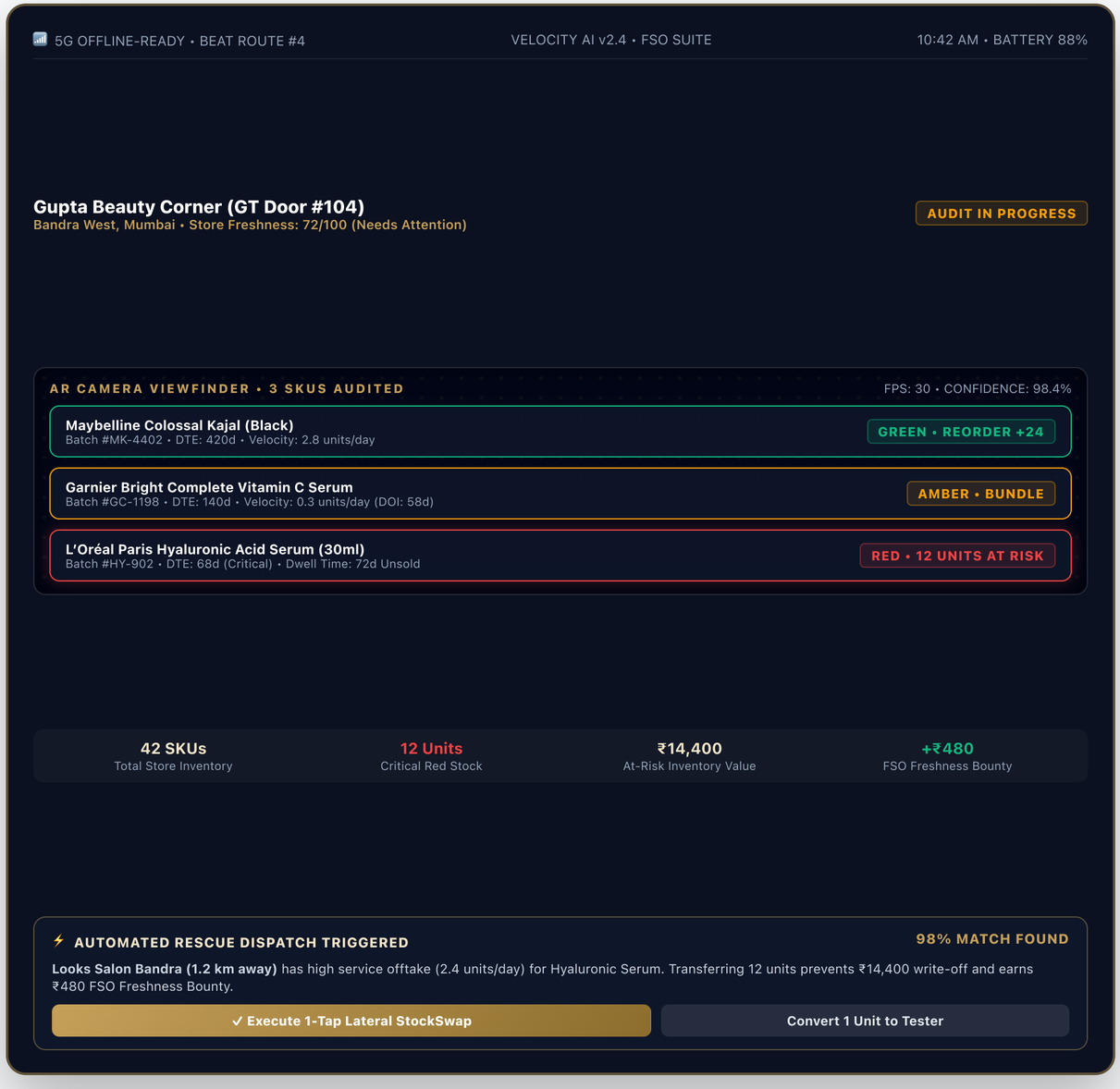
3 **Deterministic Tri-Band Triage**
Synthesizes store-level offtake velocity with regional distributor billing to classify every single SKU into three deterministic operational risk bands with prescribed action protocols.

Vision Model	Inference Speed	OCR Accuracy	Edge Database
YOLOv8n-int8 (4.2 MB)	**85 ms (On-Device NPU)**	**99.2% (GS1 Standard)**	**100% Offline SQLite**

TRI-BAND THRESHOLD & PROTOCOL MATRIX

DECISION TREE

🟢 Green Band (Fast Velocity)	Standard Replenishment Order Cycle	> 180d Remaining (DOI ≤ 21d)
🟡 Amber Band (Slow-Moving)	Planogram Re-anchoring & B2B Bundling	60–180d Shelf Life (DOI > 42d)
🔴 Red Band (Critical Expiry)	Hyperlocal StockSwap / Tester Conversion	≤ 90d to Expiry (or > 60d Stagnant)



END-TO-END TELEMETRY **Continuous Edge Synchronization:** Camera Viewfinder (30 fps) → On-Device NPU (YOLOv8n-int8) → Local SQLite Ledger → MQTT Sync → SAP S/4HANA DMS

● ZERO CLOUD LATENCY

Actionable Interventions: Amber Demand Stimulation vs. Red Circular Rescue

Differentiated playbooks for slow-moving vs near-expiry stock, backed by the Lateral StockSwap Engine (Deliverable B)

OPERATIONS & ESG ROI

Amber Playbook: Slow-Moving Stock (> 42d DOI)

DEMAND STIMULATION

- **Planogram Eye-Level Re-Anchoring:** Guided AR prompts FSO to relocate stagnant shade from bottom shelves to the Golden Eye Zone (1.2m–1.5m), directly adjacent to high-velocity Hero SKUs to capture footfall visual attention.
- **Dynamic B2B Margin Bundling:** SFA app automatically generates margin-sweetened trade bundles (e.g. *"Buy 6 Maybelline Colossal Kajals, get 2 Amber Serums at +6% retailer gross margin"*) to accelerate sell-in.
- **Consumer Trial Micro-Incentive:** Retailers unlock instantaneous ₹20 UPI scan-pay rebates for customers buying Amber-flagged items during weekend rush hours.

Trigger Metric	Margin Delta	Target Turn
Untouched > 30d	+6% B2B Gross Margin	< 14 Days Clearance

Red Playbook: Critical Expiry (≤ 90 Days Remaining)

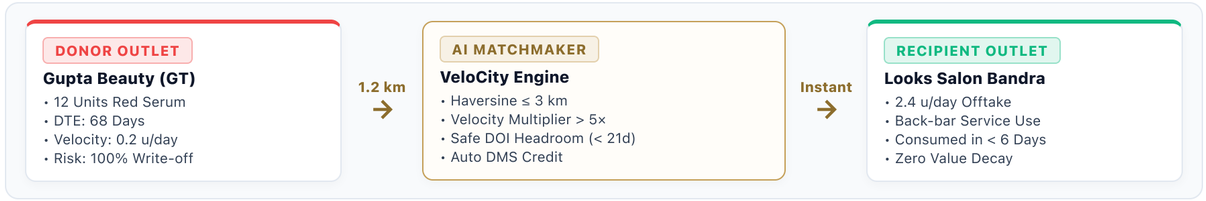
CIRCULAR REDISTRIBUTION

- **Hyperlocal Lateral 'StockSwap':** Autonomous cross-door inventory transfer. Red stock is rerouted to high-velocity salons or modern trade doors within 3 km instead of undergoing depot return.
- **Professional Salon Back-Bar Offtake:** Unsold premium hair care, treatments, and serums are liquidated at cost-neutral rates for professional in-salon services (consumed within 7–14 days).
- **Zero-Landfill Tester Unit Conversion:** 1-tap digital authorization converts 1 sealed ageing unit into an authorized in-store tester, eliminating customer purchase friction while zeroing disposal overhead.

Trigger Metric	Salvage Rate	Landfill Bleed
DTE ≤ 90d (or > 60d dead)	100% Value Recovery	0% Incinerated Scrap

HYPERLOCAL LATERAL STOCKSWAP ARCHITECTURE

ALGORITHMIC RE-ROUTING



Geographic Radius	Velocity Differential	DOI Safety Cap
Restricted to ≤ 3.0 km radius (FSO beat proximity or quick courier).	Recipient door must turn stock > 1.5× faster than donor store.	Recipient Days-of-Inventory cannot exceed 21 days post-transfer.

₹14,400	₹450	₹320	₹15,170
Avoided Inventory Write-Off	Avoided Return Freight	Avoided EPR Incineration Fee	Net Protected P&L Value

Automated Ledger Settlement: Immediate Distributor Management System (DMS) secondary credit note issued to Gupta Beauty. Zero dispute, zero manual paperwork.

100% AUDITABLE

73% Cut	₹42 Cr+	120 MT	4.5 Months
Reduction in unsellable retail returns across pilot doors	Annualized protected EBITDA margin from avoided destruction	Virgin plastic and glass packaging saved from regional landfill	Full payback period on regional edge-CV software deployment